October 21, 2026

The Editor-in-Chief
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Dear Editor,

I submit the manuscript *Certified reliability for any routing solver: screening buffered plans against travel-time uncertainty, with evidence from real delivery routes* for consideration as a research article.

Delivery firms promise time windows, but most plan with fixed travel times and protect themselves by padding every trip, which says nothing about how often the plan will be on time. Methods that model the uncertainty directly need a solver built for that model. The manuscript takes the systems view of a firm that keeps its existing solver and adds a reliability layer around it: the solver produces plans with capped internal buffers, the plans are screened by simulation, and the chosen plan is certified on independent simulated days.

The contributions are:

- a certificate that holds for any solver and selection rule (Theorem 1), together with two results on when it survives a wrong delay model: it carries over to any real delays no worse than modelled (Proposition 1), and certifying against a set of models guarantees each of them (Proposition 2);
- evidence that this works: none of 666 certified plans failed its certificate on fresh days, and when certificates under one model failed under harsher delays, certifying against five models removed every failure at a cost of about 0.65 vehicles per instance;
- a cost-of-reliability comparison with padding: a plan reaching 95% reliability existed on 46 of 53 instances with buffers and on 23 with any padding up to 50%, and used fewer vehicles;
- tests beyond synthetic data, on benchmark instances and on 40 real Amazon delivery routes with delays fitted to 6.2 million New York taxi trips.

The closest work is the robust data-driven routing of Zhang, Zhang, Lim and Sim (*Operations Research*, 2021), which also reduces to deadlines brought forward by a buffer. Their bound holds over a Wasserstein ambiguity set inside a dedicated solver; ours is a finite-sample certificate that needs no change to the solver. Simheuristics rank solutions by simulated reliability within their own search; our certificate can be added on top of any of them. These differences are set out in Section 2.

I believe the paper fits the journal's interest in operations research for logistics and in reliability. The main claims were tested in three confirmatory studies whose hypotheses were fixed and publicly timestamped before they were run, and all code, instances and records are openly available. The limitations are stated plainly. On the real routes, with the HGS solver, screening was more reliable than padding by 8.0 and 10.1 points under two of the three delay models, at the same number of vehicles; under the mildest model padding was already reliable and slightly ahead.

The manuscript is not under consideration elsewhere and has not been published. Generative AI was used for code, analysis, literature search and drafting, as disclosed in the manuscript; the author takes full responsibility for the content. There are no competing interests and no funding to declare.

Suggested reviewers, none of whom has any connection with the author:

- Prof. Jan Fabian Ehmke, Department of Business Decisions and Analytics, University of Vienna (jan.ehmke@univie.ac.at), for service levels in routing with stochastic travel times;
- Dr. Ola Jabali, Operations Research Group, DEIB, Politecnico di Milano, for self-imposed time windows and routing under uncertainty;

- Prof. Angel A. Juan, Universitat Politècnica de València, for simheuristics.

Thank you for considering the manuscript.

Yours sincerely,

Suhan Ali
Independent researcher, Lahore, Pakistan
syedsuhanali9@gmail.com ORCID 0009-0004-5079-7684